

Math Fluency through Game Design

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Abstract. Our goal in this research is to create a comprehensive framework establishing guidelines for the design of math fluency games for adult learners. Our user-centered design approach consisted of focus groups with students, faculty, and administrators from a two-year and a four-year institution to probe more deeply into the ways students perceive the value of math in everyday activities. Using our comprehensive focus group protocol, we evaluated users' perception and understanding of culture-based mathematics to determine value-laden game designs that will promote math fluency among developmental math students. During these sessions, we collected quantitative and qualitative data in the form of survey data, play-test data, and field notes. The data speak to various issues such as games as a learning tool, interests and mismatches between designers and the target audience. Moving forward, our research will provide future directions for defining holistic usability by integrating user-centered design and game design.

Keywords: Game Design, User Centered Design.

1 Introduction

People who are math literate have knowledge or competence of basic math skills, while people who are math fluent having the ability to unconsciously and smoothly demonstrate mastery. Even though math is integrated into many aspects of everyday life, many people are math literate, but not math fluent [1]. The majority of our society is capable of appropriating math in terms of telling time, adding single digit numbers, etc. However, many struggle with applying mathematical foundations, principles, and theories such as balancing a checkbook, or calculating a tip. According to Ginsburg et al. [2], 58.6% of U.S. adults were below level 3, the minimum level for coping with today's math skill demands. This may seem trivial in this technological age where everyone has immediate access to calculators or other technological devices. However, lacking in these skills can be extremely problematic when those same skill sets become the barrier to successfully meeting goals in life.

2 Background

In order to strengthen scientific competitiveness in the U.S., there has been a rapid societal shift from a need for laborers to a need for knowledge workers [3]. In parallel with this shift, the recession is forcing many adults (ages 25-54) to return to higher education, such as trade schools and community colleges, in order to develop their skill sets and expand their career options. For many of these learners, however, a lack of math skills, necessary for most entry-level community college curricula, will prevent them from progressing toward their goals. In fact, while sixty percent of all community college students are required to take at least one developmental math course to matriculate, only thirty-percent pass the developmental math courses they need [4]. This means that forty-two percent of all community college students are failing to pass the math they need for their coursework. This lack of preparation in basic math skills, paired with the hierarchical nature of math, results in hindered comprehension of more advanced mathematical skills [5]. To address this issue, building math fluency -- the ability to perform a behavior correctly, quickly, and with minimal effort [6]-- is key. Math fluency contributes to learning outcomes of performing a skill or recall knowledge independently, maintain performance levels, and effectively reappropriate that same skill and knowledge [7]. Educational games are one approach for building math fluency because of what they offer in terms of engagement, repetitive actions, immersion, and exposure time [8]. While games might not immediately seem like a viable approach for adult learners, a recent Pew Internet survey [9] found that 53% of American adults participate in digital games through mobile applications, video games, or electronic games. Though games are a popular medium, not all types of games are popular with all audiences. For example, women tend to be drawn more towards cooperative types of games [10] versus competitive games and older adults prefer games on the computer as opposed to console and mobile games [9]. As with any software, a keen understanding of the target audience is needed to build effective math fluency games for adult learners. However, there are few guidelines for designing math educational games targeted towards this unique subset of users. We propose to use user-centered design [11] techniques to create a design framework that establishes guidelines and criteria for math fluency games for adult learners. To do so, we must build an explicit understanding of users, tasks, and environments; and this knowledge must be iteratively refined during the design and development process [12].

3 Method

The overall project's goal is to improve performance in developmental mathematics courses by building games that help students develop the number sense required to succeed in mathematical problem solving. To improve our designs for this specific audience, we evaluated students' perception of their learning of math and how they think about learning math from game environments. This study was conducted in three parts. Each part of the study focused on a different aspect of the design to better

understand the target audience. Part I was an online pre-questionnaire/survey on basic demographic information. Part II was a focus group discussion where participants describe and identify math in their everyday activities. The focus group protocol used was an adaption of the Family Math Protocol created by the Family Math Project [13], with questions designed to solicit narratives about math usage in everyday activities. Part III consisted of play-testing one or two math-centered educational games, where students played games and provided their feedback. After each game they completed an evaluation and provided their own ideas of what an ideal math learning game would entail. The data of part III is beyond the scope of this paper as we are focusing primarily on students' perception of their learning process.

3.1 Participants

We recruited participants from the STARS Computing Corps program at Central Piedmont Community College (CPCC), and the Fall 2013 developmental math courses at University of North Carolina-Charlotte (UNCC) through in-class announcements. The STARS Computing Corps seeks to increase the participation of women, underrepresented minorities, and persons with disabilities in computing disciplines through multi-faceted interventions that lead to computing careers. These STARS students at CPCC had already completed higher-level math and computing courses, and participated in projects where they applied these skills. As such, the CPCC students are considered to have high math fluency (HMF). Conversely, we consider the UNCC students to have low math fluency (LMF) because they were recruited from the lower level developmental courses.

3.2 Procedure

At the start of the study, participants were asked to read and sign the consent form. Then, participants completed the pre-questionnaire on demographic information and prior education. The questionnaire gives the researcher a better feel of the group in preparation for the focus group. Discussion occurred in four sessions. During the discussion participants were asked to think and respond to each question as the moderator asks them. The discussions lasted for about 30-45 minutes. Studies took place on the students' respective campus, in Charlotte, North Carolina. Participants were compensated with gift cards for participating in the study.

4 Results

In total, 17 participants participated in the entire study with a gender breakdown of (n=7) male participants and (n=10) female. Students were majoring in both Science-related fields (n=11) and the Liberal Arts (n=6). Table 1 further describes the demographics of the participants. We provide this level of detail since a thorough understanding of the user is fundamental to user-centered design.

Table 1. Description of Participants in Study, with numbers (and percentages) given for each group

Demographics		n(%)	
		CPCC	UNCC
Gender			
	Male	8 (88.9)	1(10.0)
	Female	1(11.1)	9(90.0)
Highest Education completed			
	High school diploma (including equivalency)	1(11.1)	4(40.0)
	Some college, no degree	7(77.8)	6(660.0)
	Associate degree	1(11.1)	
Age			
	18-24 years	4 (44.4)	10(100)
	25-34 years	4(44.4)	
	35-44 years	1(11.1)	
Ethnicity			
	White or Caucasian	4 (44.4)	3(30.0)
	Black or African American	3(33.3)	3(30.0)
	Hispanic or Latino	1(11.1)	
	Asian	1(11.1)	
	Multiracial/ Other (please specify		4(40.0)

4.1 High Math Fluency - STARS students at CPCC

The HMF focus group consisted of (7) males and (1) female. Majors were primarily computer science-related (7 of 9 participants). Most reported having a high math fluency (HMF) and had positive historical experience with math learning and activities. The most common hobbies reported were going to the gym/exercising, programming, and reading. Most participants reported a moderate-to-high interest in play videogames and one participant had experience with programming rudimentary games. Most participants expressed a comfortable working knowledge of math or even an appreciation of using math; “for me personally, I love puzzles and that's essentially what math is; it's...just concrete evidence and just proof and facts and it just like is everything of what math is... I wouldn't believe anything without...” (HMF, P2) “...Proof.” (P4) “Right, exactly” (P2). When participants described how math could be seen in everyday life, the most common responses were angles, force and trajectory in sports such as basketball, swimming, billiards and soccer. Participants agreed that in order to achieve a high level of play, it was necessary to consider the math involved in athletic competition, “...in basketball, if you don't see

the angles you're not going to be good.” (HMF, P4). Another participant used the recent Olympics as an example, “with the Olympics this past year...they had swimmers putting...sensors on their bodies so they could see how...the drag...in the water [could] be reduced” (HMF, P5). Although they acknowledged this need for math in sports, participants in this group unanimously agreed that focusing too much on the math involved would have a negative impact on the level of fun. “When you break it down to it's mathematical form you've removed all passion out of it. Who's gonna sit there and think while they're playing a little game about the physics of it?” (HMF, P1) Though several participants in this group said they enjoy playing video games, most were skeptical about the value of games as an educational tool for adult learners such as themselves: “For me it was learning by touching and doing. It wasn't learning by playing. Not everything can be fun, and that's something that also needs to be learned” (HMF, P5).

4.2 Low Math Fluency –Developmental Math students at UNCC

The LMF groups consisted of a focus group with (7) participants and (2) in-depth interviews with (1) participant in each interview. All participants from LMF were female and majors were split almost even with (4) students from Science-related fields and (5) from Liberal Arts fields. The most common hobbies reported were reading, listening to music, watching TV/movies, religious activities and playing sports. The most frequently mentioned use of math in everyday life was budgeting/financial management. The second most common was time management. “I use math when I'm deciding on what time I should wake up in the morning...I set my alarm clock to the last possible minute. I get ready in like twenty minutes.” (LMF, P7) Overall, the LMF participants made more references to friends and family when discussing hobbies, learning and playing games. “I'm beating my mom [at Words with Friends]...I have like 230, she has like 170, and I think I'm beating my other friend by maybe 20 points...” (LMF, P9) Overall, participants spoke extensively of low math fluency (LMF), claiming to have inferior math skills or possess a strong dislike for math in general with (19) negative references and only (4) positive references in regard to their thoughts on math. Two participants mentioned being capable with math, “*I was naturally good at math, but with Physics I was actually able to see how it works*” (LMF, P6), but others in this group spoke negatively of their math skills and admitted to struggling with the subject, “*math is...my weakness and it has always been*” (LMF, P3). Their opinion of the subject seemed directly influenced by the perception of their abilities, “*I don't like math, like simple math is okay, but I don't do well a whole lot of math.... If I'm good at something, I like it. If I'm bad at something, I don't like it.*” (LMF, P1).

Having problems broken down seemed key to understanding for some participants, “*...the easiest way to learn it would be to have it... broken down...*” (LMF, P2). Participants also stressed their need for step-by-step instructions; “*I probably learn the best when the steps are in detail. So, like, 'step one is this, step two is this, and then three and then four', however many...that really helps me, to go step-by-step.*” (LMF, P8).

Nearly all participants from the LMF group spoke of getting frustrated when teachers or others would falsely assume the participant understood a particular concept; *"I never learned this. Can you explain this formula to me?" and they're just like, 'you should have learned this two years ago. It'll be okay'"* (LMF, P5). *"I feel like a lot of the teachers I have... they think that you should know it, so they just keep on going and they don't stop"* (LMF, P1). When participants in this group did not have their needs addressed when learning math, they spoke of withdrawing completely, *"I'm just like 'I don't know', and...I get frustrated...I'll just be like 'forget it'"* (UNCC, P7) or giving up on trying to learn, *"I just got to the point where... I just couldn't study it because... I was like, I don't know this so what's the point of me studying it. I'm not going to get it...I turn all the way off. I shut down. I start ignoring it."* (LMF, P3). Instead of seeking assistance, participants spoke of internalizing their failure, *"If I don't know what I'm doing it's, it's really hard for me to ask for help 'cause I feel stupid"* (LMF, P1). Participants spoke positively about previous math education games played in Elementary and Middle School, for example, The Lemonade Stand game, but had mixed opinions of games being used in their current math courses, *"I can't honestly...really picture it. I think it'd be too juvenile...I can't picture...a math game with the math we're doing now. I feel like I'd just get bored with it."* (LMF, P6) Half of the participants believed the addition of games would be beneficial and said they would be open to the idea, *"I think the math games would work, because I think one thing why math is so unpopular is because you actually have to put time into understanding things and I think games provides it in like a fun way...like you're not dragging on..."* (LMF, P7) *"...doing homework problems out of a textbook for two and a half hours..."* (LMF, P5). The other participants said games would not be effective for them, citing concerns such as the inability to ask questions, the idea that games might be "too juvenile", and having to stop and do problems with pen-and-paper would be similar to homework and therefore detract from the level of fun, *"...you'd be like, 'okay, this is gonna be awesome and then you'd get to the actual mini games and it's like, this sucks, this is just school all over again'"* (LMF, P2). One participant did not comment on the topic.

4.3 Cross-Group

When asked, "How often do you play digital games (online, on your phone, on the computer, console, etc)", about half of the participants overall reported playing some type of digital game often. The two groups differed, however, in the types of games they played. The HMF students primarily played console and first person shooter games, while the LMF students primarily described games played on the mobile games, cooperative games, etc. For example, one student from HMF mentioned Black Ops and one student from LMF mentioned the Sims 3. Participants in both groups expressed an interest in the repetition of fundamental skills, with an HMF participant admitting love of math started at an early age: *"I remember when I was younger, my father would do drills with me (laughs) and he would make me do drills over and over..."* (HMF, P3) One LMF participant remembers slowly building confidence, *"once I get it right, I'll repeat it over and over and over again. and then...that's when*

I'll kind of start trying to figure out the shortcuts and then I repeat it and that's kinda how I build... I'll use a bunch of little math and I know it takes longer, but once I get that, that's when I kind of build off it, but it's mostly repetition for me.” (LMF, P1)

5 Analysis

5.1 High Math Fluency - STARS Students at CPCC

The HMF students saw math as ubiquitous. As they described how one would come to excel in some of their hobbies and activities, they would often discuss it in terms of the higher-level math it entailed; for example, algorithms in video games and angles in basketball, billiards, etc. When asked about how they approached math, the HMF students frequently responded, “we just do it.” This casual sentiment displays fluency, their ability to smoothly navigate through much of their day-to-day math skills without much thought. In addition, the average age of the HMF students was older than the LMF students, which could impact their opinion on games used as learning tools. Many spoke of learning math fundamentals prior to the heavy use of technology, such as calculators, in the classroom. Additionally, the types of games participants cited were typically complex, high graphic console games such as Black Ops, etc. This could impact their opinion of games as a learning tool as their expectation of the constitution of games would be more than simple 2D graphics.

5.2 Low Math Fluency –Developmental Math Students at UNCC

The LMF students often lack the confidence in their ability to fully grasp the math used in everyday activities independently, for example calculating discounts while shopping. They identified simple direct math existing in various activities such as tipping, measuring, and counting. However, they often do not possess the comprehensive understanding behind tasks that call for more complex math. For example, one participant described ice-skating and using trial and error to effectively make turns rather than explaining the force, trajectory, and angles that are inherent in the process.

Participants involved in the study seemed to value community as participants made more references to friends and family when discussing hobbies and more reference to one another - even at times finishing each other’s sentences. Most hobbies included descriptions of other people in addition to the participant - for example: cooking with family, team sports, traveling with friends/family, and playing games with friends/family. When responding to questions, LMF used more high-context style descriptions than participants in the HMF group. The types of games played primarily consisted of simple, community or collaborative-style games such as Words with Friends and The Sims 3. Participants in the study also seemed highly influenced by external perceptions by describing situations in which the teachers’ attitude negatively impacted the students’ perceptions of their abilities. The negative perceptions of their ability to learn math may make them both less likely to seek help when unable to understand, and quicker to lose interest. Essentially for these LMF students, poor

performance in math becomes a self-fulfilling prophecy. When learning a new concept, or reviewing previous concepts, participants in this group preferred to have problems broken down into step-by-step instructions or easily-digestible components. The LMF students frequently emphasized the need to visualize and understand the entire problem-solving process as it is developing. This step-by-step method was seen as a means to providing building blocks and build confidence and competence with math skills. When imagining math learning games, some of these students seem to feel that they would be superficial and juvenile exercises that required students to do math outside the game in order to play.

5.3 Cross-Group

About half of all of the participants played digital games regularly. While their perception of how they engaged in the games differed greatly, both groups were opposed to learning that impeded the fun-factor in the realm of playing games. Participants equated this to parents hiding vegetables in food - they can get away with it when children are young, but after a certain age, we all know they are there. One of the most important themes pulled from this research was that individuals learn math at different speeds and through different methods. Some participants preferred visual tactics while others emphasized learning by doing, but everyone enjoyed the positive feeling that came from understanding and feeling successful. Participants stated the need for repetition of fundamentals/basics in order to move on to a higher level or more complex math. Participants across both groups described using math every day through activities such as time management and budgeting. As all participants are current students, time management concerns revolved around balancing schoolwork and course schedules with extracurricular activities, hobbies, and sleeping. Financial management consisted of budgeting; the primary budgeting mentioned involved food and shopping in general.

6 Recommendation

Due to the variety and depth of information gathered, we feel valuable considerations may be drawn for future game design research. When designing developmental games for the audience of LMF students, the following should be taken into consideration:

- Due to the variety and depth of information gathered, we feel valuable considerations may be drawn for future game design research. When designing developmental games for the audience of LMF students, the following should be taken into consideration:
- The design of the game should include options for several learning styles, e.g., auditory, visual, and kinesthetic.
- Games should allow players to set their own pace or should adapt to each individual's pace of learning.

- Players are more likely to disengage faster when they don't understand the math or how to play.
- Players may be less likely to seek help or assistance when they feel they possess inadequate math skills. Therefore, the game should scaffold problem solving with built-in step-by-step instructions. This may downplay the negative perception of needing assistance.
- Design for multi-player and/or collaborative game experiences..
- Do not try to hide the math. Instead, incorporate high-context scenarios that are applicable to the target audience outside of the game.
- Provide positive, reinforcing feedback that allows students to perceive incremental improvements to build confidence.
- Embed the math within the game as the core game mechanic, so students don't have to work problems on paper outside of the game environment, or feel that the game is superfluous to learning.
- Explicit connection should be made between the game, the skill, and the context it will be applied in outside the game as LMF students do not see how games would facilitate practice of the skills they need in everyday life.

7 Conclusion

The aim of this paper is to better inform the design of games targeted to increase the math fluency of developmental math collegiate students. This was accomplished by exploring how developmental math students perceive learning math overall and how they think about learning math from gaming environments. Using focus groups and in-depth interviews, we attempted to probe more deeply into the ways students perceive the value of math in their everyday activities and connect these experiences to their learning of math. The main findings from this research emphasized that, while participants have different needs when it comes to learning math, most emphasized the need for step-by-step instructions. With this, low math fluency students feel that fully understanding basic math is essential before adding a higher level or more complex math. Repetition of the fundamentals is the best way to gain this comprehensive understanding for these participants. Additionally, multi-player collaborative games can prove quite beneficial to low math fluency learners as it depicts their preferred learning and gaming environment of being part of a community. Based upon this information, we presented several recommendations for designing educational games to help adults build math fluency. Respecting the needs and perspectives of adults within the designs of these games is critical for their potential adoption. Games designed without sufficient scaffolding or instruction can leave this audience frustrated and uninterested. Instead of being tricked to develop math skills in abstract game contexts, these low math fluency adults want to know specifically how and when they will use the skill they are developing.

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